

## Assignment 2: Max Warriner (126104460)

1.

a. Find in R Script

b.

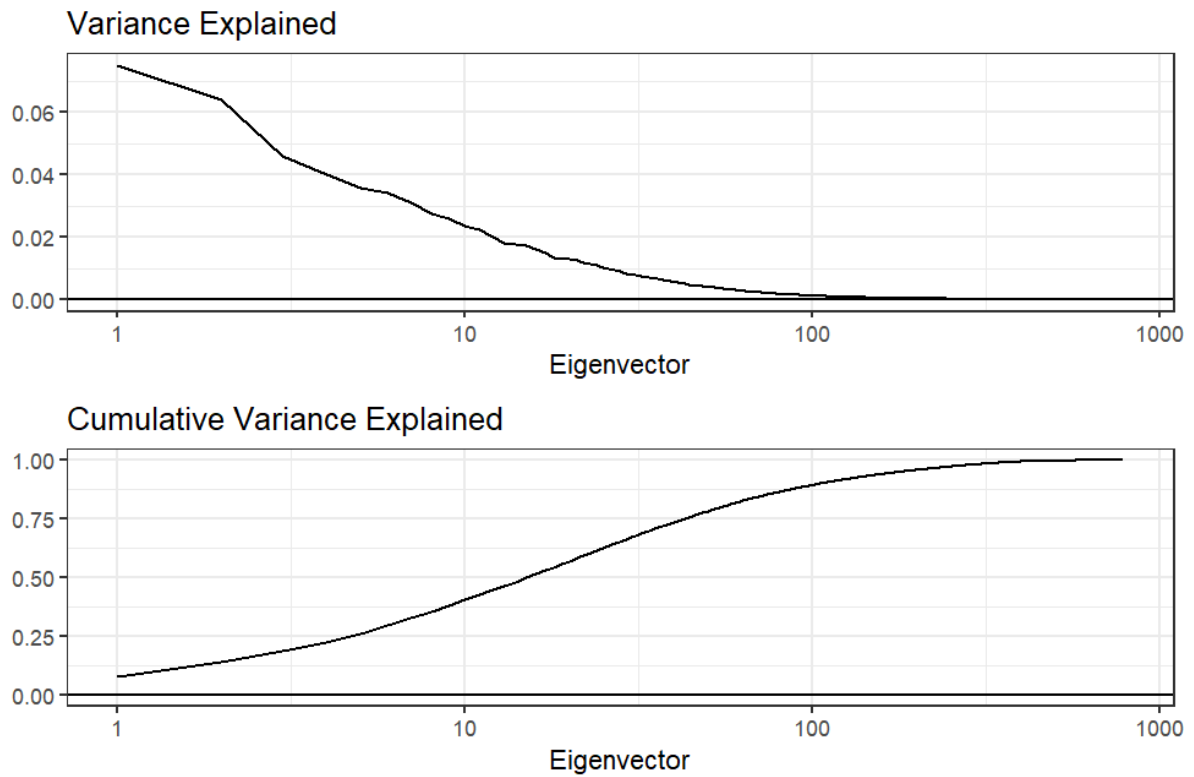


Figure 1. Eigenvalue Variance Analysis

346 eigenvectors are needed to explain at least 99% of the variance in the data.

c. Find data structure in R Script

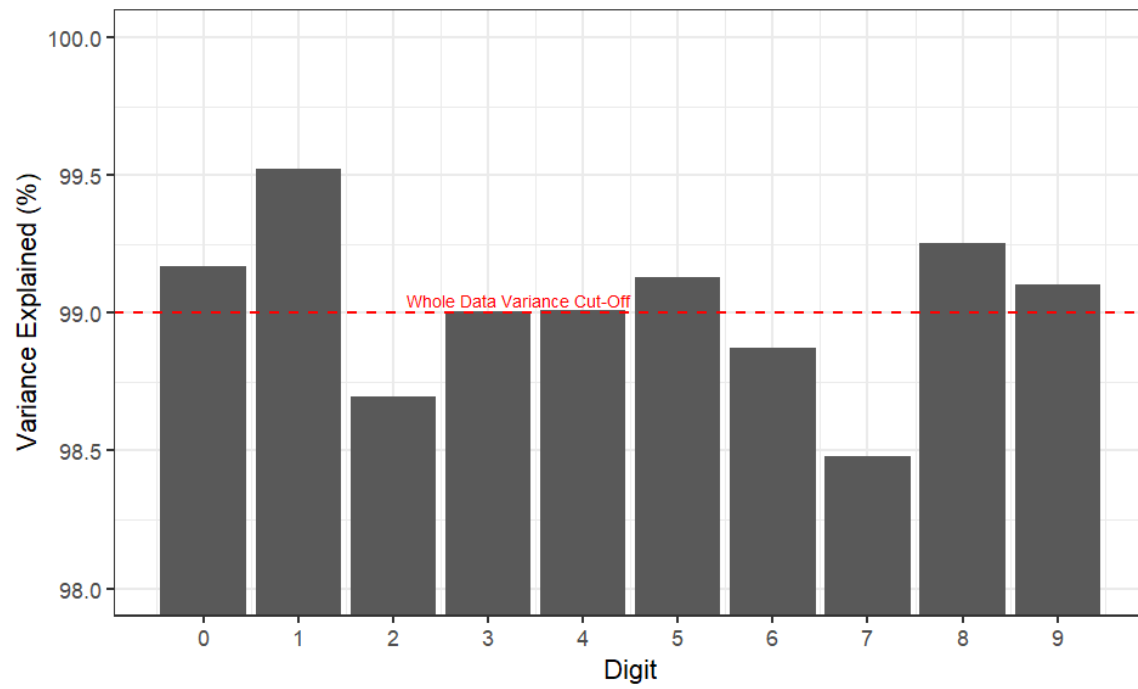


Figure 2. Digit Variance Analysis

2.

- Find in R Script
- (also c.)

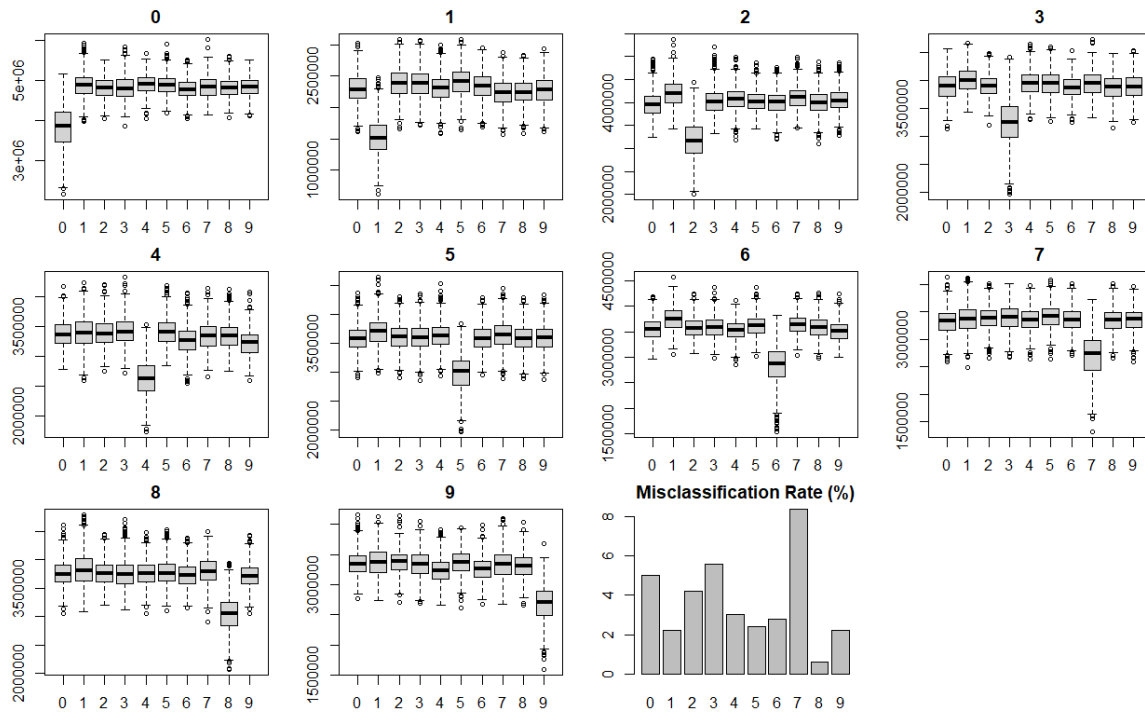


Figure 3. Residual Sum of Squares by Digit Analysis

3.

- Number of Stars Removed in Each Class

| Star Type | Outliers Removed |
|-----------|------------------|
| BYDra     | 217              |
| CEP       | 0                |
| CEPII     | 0                |
| DSCT      | 0                |
| EA        | 45               |
| EW        | 75               |
| Mira      | 2211             |
| RR        | 1                |
| RRc       | 0                |
| RSCVN     | 134              |
| SR        | 5134             |

b.

i.

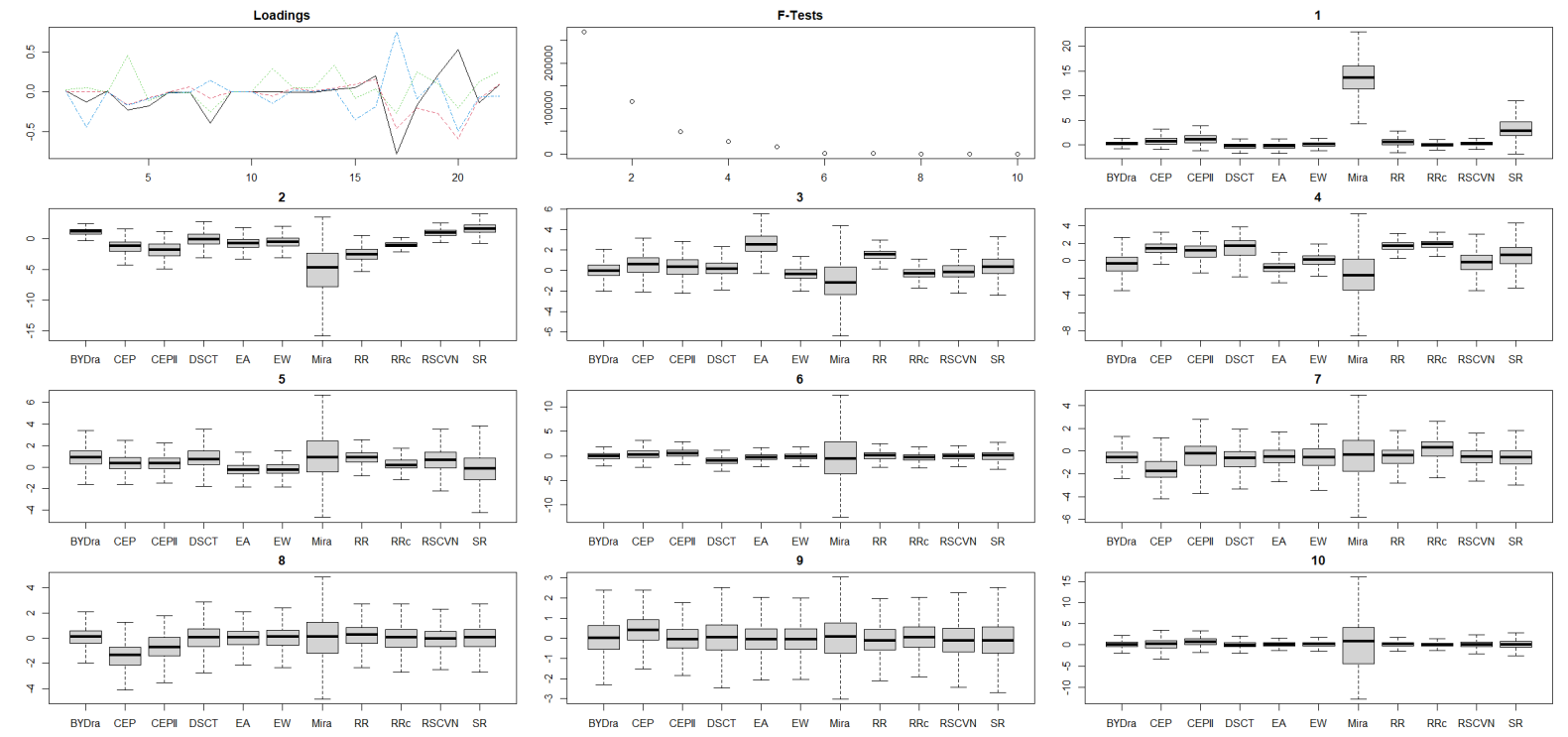


Figure 4. LDA Analysis of Stars Data

ii.

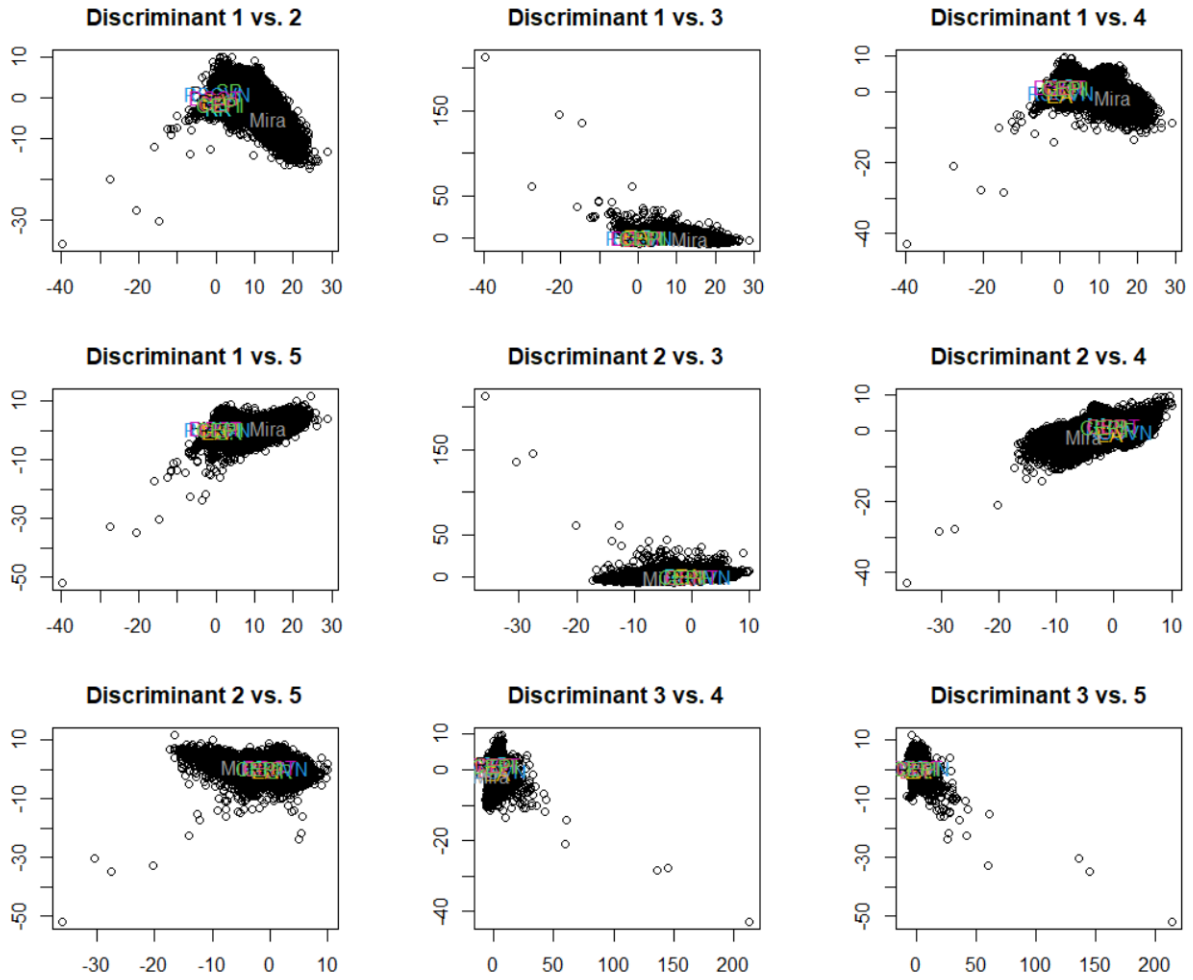


Figure 5. Pairwise Discriminant Analysis of Stars Data

c.

i. Linear Discriminant Analysis Misclassification Rate Analysis

| Star Type | Misclassification Rate |
|-----------|------------------------|
| BYDra     | 44.1%                  |
| CEP       | 85.0%                  |
| CEPII     | 98.9%                  |
| DSCT      | 75.5%                  |
| EA        | 23.1%                  |
| EW        | 6.4%                   |
| Mira      | 9.1%                   |
| RR        | 17.9%                  |
| RRe       | 88.7%                  |

|       |       |
|-------|-------|
| RSCVN | 87.7% |
| SR    | 26.9% |

#### Quadratic Discriminant Analysis Misclassification Rate Analysis

| Star Type | Misclassification Rate |
|-----------|------------------------|
| BYDra     | 27.0%                  |
| CEP       | 30.9%                  |
| CEPII     | 59.2%                  |
| DSCT      | 5.2%                   |
| EA        | 18.0%                  |
| EW        | 19.5%                  |
| Mira      | 0.5%                   |
| RR        | 4.0%                   |
| RRc       | 8.4%                   |
| RSCVN     | 73.9%                  |
| SR        | 7.9%                   |

#### ii. Overall Misclassification Rates

Linear Discriminant Analysis Misclassification Rate = 26.8%

Quadratic Discriminant Analysis Misclassification Rate = 22.9%

We see that overall, the quadratic discriminant analysis performs better than the linear discriminant analysis, showing evidence for different covariance structures between different star types that is useful for classification.

A simple correlation analysis shows that sample size has a strong negative correlation with misclassification rate in the LDA ( $r = -0.49$ ), but less so in the QDA ( $r = -0.01$ ). This makes sense with the poorer performance in the LDA, where the model tends to overestimate the larger classes as opposed to the more sensitive and precise QDA analysis.